Systems of rational equations

Two rational equations at once — substitution, symmetric sums, and the domain that must survive both.

LESSON 30–31 | 2 × 40 MIN | ALGEBRA 10, §2.2 | P1 · 1.7

LESSON CLOCK

8' | 20' | 22' | 16' | 20' | 4

What is new	8 min
Substitution	20 min
Symmetric systems	22 min
Graphical meaning	16 min
Practice	20 min
Homework	4 min

LEARNING OBJECTIVES

- Solve a system containing rational equations by substitution.
- Use the sum-and-product substitution for symmetric systems.
- State the combined domain of a system.
- Interpret the solutions as points of intersection.

TEXTBOOK

National	pp. 104–113
Cambridge	Pure Mathematics 1, pp. 18–23
Workbook	P1 Exercise 1F

01 Explanation

The combined domain

BOTH EQUATIONS MUST ACCEPT THE ANSWER

A pair (x, y) solves the system only if it lies in the domain of **both** equations. So the excluded values of each are excluded from the whole system.

Everything else is the ordinary technique of Grade 9 — eliminate a variable, solve, substitute back — with the extra step of clearing denominators first.

Substitution

When one equation gives a variable cheaply, use it:

$$x + y = 5 \text{ and } \frac{1}{x} + \frac{1}{y} = \frac{5}{6}$$

The second equation is $\frac{x + y}{xy} = \frac{5}{6}$, so $\frac{5}{xy} = \frac{5}{6}$ and $xy = 6$. With $x + y = 5$ the numbers are the roots of $t^2 - 5t + 6 = 0$.

$$(x, y) = (2, 3) \text{ or } (3, 2)$$

TWO ORDERED PAIRS, NOT ONE

A symmetric system almost always gives both orders. Writing only one loses half the answer.

Symmetric systems

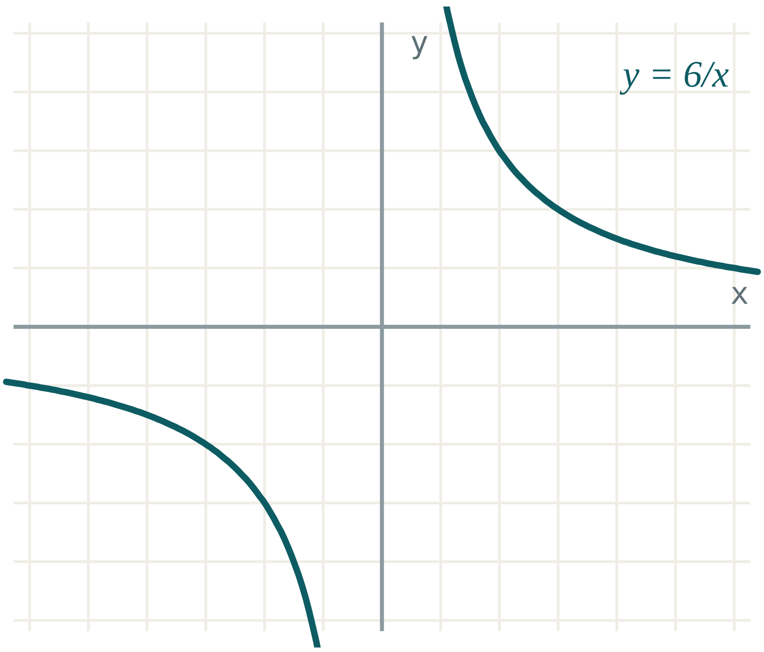
A system is **symmetric** when swapping x and y leaves it unchanged. Every such system can be rewritten in $s = x + y$ and $p = xy$:

EXPRESSION	IN S AND P
$x^2 + y^2$	$s^2 - 2p$
$\frac{1}{x} + \frac{1}{y}$	$\frac{s}{p}$
$x^3 + y^3$	$s^3 - 3ps$
$(x - y)^2$	$s^2 - 4p$

Solve for s and p, then recover x and y as the roots of $t^2 - st + p = 0$.

What the solutions mean

Each equation is a curve. The solutions of the system are the points where the curves cross — so the number of solutions is the number of intersections.



xy = 6 is a hyperbola; x + y = 5 is a line. Two crossings, two solutions.

A system with no solution is a line missing its curve entirely; a system with one solution is a tangency.

02 Worked examples

EXAMPLE 1 **Solve $x + y = 7,\ \frac{1}{x} + \frac{1}{y} = \frac{7}{12}$.**

1. $\frac{x + y}{xy} = \frac{7}{12}$
2. $\frac{7}{xy} = \frac{7}{12} \Rightarrow xy = 12$
3. $t^2 - 7t + 12 = 0$
4. $t = 3, 4$

ANSWER

$(3, 4)$ and $(4, 3)$

EXAMPLE 2 **Solve $xy = 12, x^2 + y^2 = 25$.**

1. $s^2 - 2p = 25$ with $p = 12$.

   $s^2 = 49$

2. $s = \pm 7$

3. $s = 7$: $t^2 - 7t + 12 = 0 \Rightarrow 3, 4$.

4. $s = -7$: $t^2 + 7t + 12 = 0 \Rightarrow -3, -4$.

ANSWER

$(3,4), (4,3), (-3,-4), (-4,-3)$

EXAMPLE 3 **Solve $\frac{2}{x} - \frac{1}{y} = 1, \frac{1}{x} + \frac{2}{y} = 3$.**

1. Let $u = \frac{1}{x}$, $v = \frac{1}{y}$.

   $2u - v = 1, u + 2v = 3$

2. Solve: $u = 1, v = 1$.

3. $x = 1, y = 1$

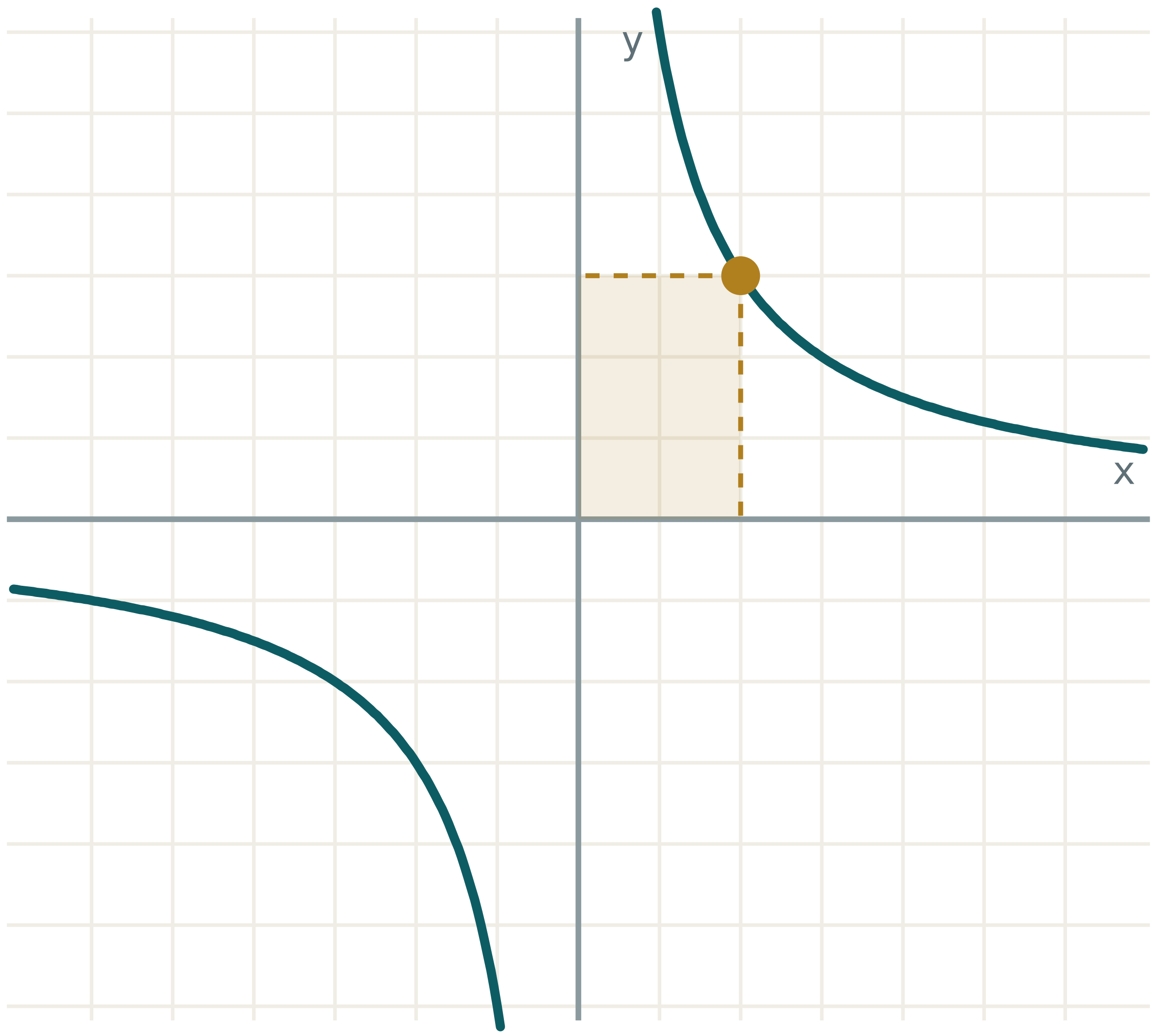
ANSWER

$(1, 1)$

03 Interactive model

Plot both curves and count the crossings before solving.

The graph of y = k ⁄ x



k **6** | x **2** | y = k/x **3** | x · y **6** | quadrants **I and III**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
System of equations	Tenglamalar sistemasi	Система уравнений
Substitution method	O‘rniga qo‘yish usuli	Метод подстановки
Symmetric system	Simmetrik sistema	Симметричная система
Sum and product	Yig‘indi va ko‘paytma	Сумма и произведение
Combined domain	Umumiy aniqlanish sohasi	Общая область определения
Point of intersection	Kesishish nuqtasi	Точка пересечения
Consistent system	Birgalikdagi sistema	Совместная система
Inconsistent system	Birgalikda bo‘lmagan sistema	Несовместная система

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A symmetric system usually gives:

- one pair
- two ordered pairs
- no pairs
- infinitely many

$x^2 + y^2$ in terms of s, p is:

- $s^2 + 2p$
- $s^2 - 2p$
- $s^2 - p$
- $2s - p$

For $\frac{1}{x} + \frac{1}{y}$ a good substitution is:

- $u = x, v = y$
- $u = \frac{1}{x}, v = \frac{1}{y}$
- $u = xy$
- none

The solutions of a system are:

- the roots of one equation
- the points where the curves meet
- the domains
- the asymptotes

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $x + y = 5, xy = 6$

ANSWER $(2,3), (3,2)$

2. Solve $x + y = 8, xy = 15$

ANSWER $(3,5), (5,3)$

3. Solve $x - y = 1, xy = 6$

ANSWER $(3,2), (-2,-3)$

4. Write $x^2 + y^2$ with s, p

ANSWER $s^2 - 2p$

5. Write $\frac{1}{x} + \frac{1}{y}$ with s, p

ANSWER $\frac{s}{p}$

6. Domain of $\frac{1}{x} + \frac{1}{y} = 1$

ANSWER $x \neq 0, y \neq 0$

7. Solve $xy = 4, x = y$

ANSWER $(2,2), (-2,-2)$

Medium · the standard question

7 PROBLEMS

1. Solve $x + y = 7, \frac{1}{x} + \frac{1}{y} = \frac{7}{12}$

   ANSWER $(3,4), (4,3)$

2. Solve $xy = 12, x^2 + y^2 = 25$

   ANSWER $(\pm3,\pm4)$ matching signs

3. Solve $\frac{2}{x} - \frac{1}{y} = 1, \frac{1}{x} + \frac{2}{y} = 3$

   ANSWER $(1, 1)$

4. Solve $x + y = 6, x^2 + y^2 = 20$

   ANSWER $(2,4), (4,2)$

5. Solve $x - y = 2, \frac{1}{x} - \frac{1}{y} = -\frac{1}{4}$

   ANSWER $(4,2)$ or $(-2,-4)$

6. Solve $xy = 6, x + y = -5$

   ANSWER $(-2,-3), (-3,-2)$

7. How many solutions has $x + y = 1, xy = 1$?

   ANSWER none — $D < 0$

Hard · stretch and reason

7 PROBLEMS

1. Solve $x + y = 4,\ x^3 + y^3 = 28$

   ANSWER $(1,3),\ (3,1)$

2. Solve $x^2 + y^2 = 13,\ \frac{1}{x} + \frac{1}{y} = \frac{5}{6}$

   ANSWER $(2,3),\ (3,2)$

3. Solve $\frac{x}{y} + \frac{y}{x} = \frac{5}{2},\ x + y = 6$

   ANSWER $(2,4),\ (4,2)$

4. Solve $x + y + xy = 11,\ x^2 y + x y^2 = 30$

   ANSWER $(2,3),\ (3,2)$ or $s = 6,\ p = 5$

5. Find k so $x + y = 4,\ xy = k$ has exactly one solution

   ANSWER $k = 4$

6. Solve $\frac{1}{x} + \frac{1}{y} = \frac{1}{2},\ \frac{1}{x^2} + \frac{1}{y^2} = \frac{5}{36}$

   ANSWER $(6,3),\ (3,6)$

7. Interpret $x + y = 5,\ xy = 6$ graphically

   ANSWER A line meeting a hyperbola in two points

07 Homework

Homework — 5 tasks

Write both ordered pairs whenever the system is symmetric.

1. Solve $x + y = 9,\ xy = 20$.
2. Solve $x + y = 5,\ \frac{1}{x} + \frac{1}{y} = \frac{5}{6}$.
3. Solve $x^2 + y^2 = 41,\ xy = 20$.

4. Solve $\frac{3}{x} + \frac{2}{y} = 4$, $\frac{1}{x} - \frac{1}{y} = 1$.

5. Explain, with a sketch, why $x + y = 1$ and $xy = 1$ have no common solution.